Software Atelier I

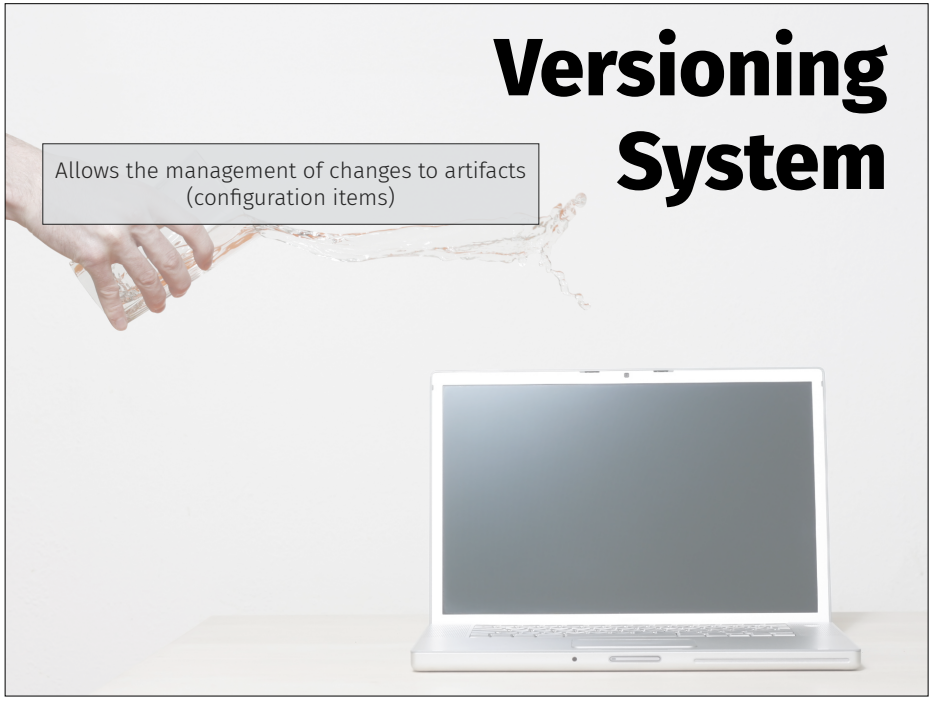
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Versioning System



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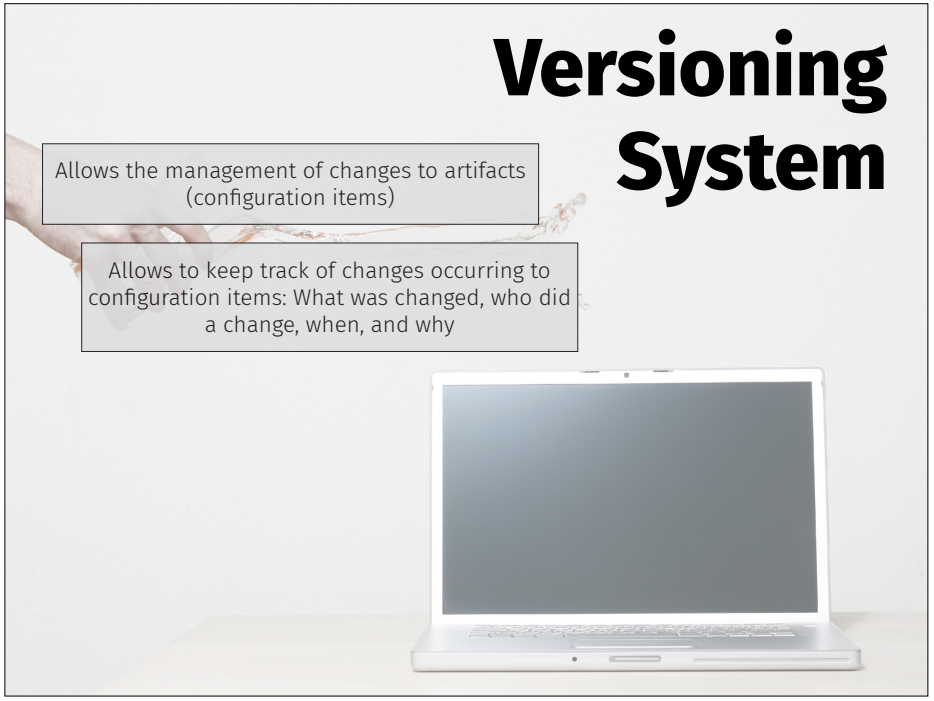
Allows the management of changes to artifacts (configuration items)



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Allows to keep track of changes occurring to configuration items: What was changed, who did a change, when, and why

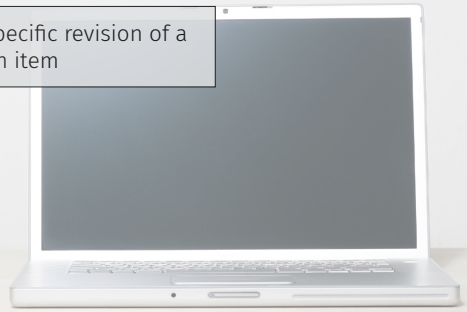


Versioning System

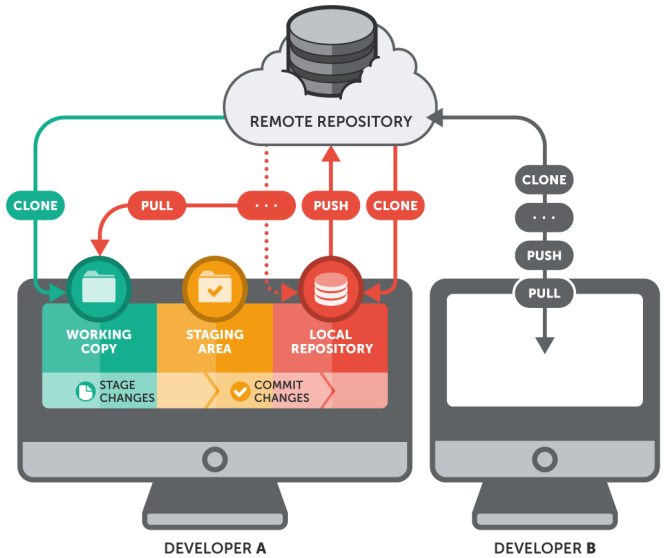
Allows the management of changes to artifacts (configuration items)

Allows to keep track of changes occurring to configuration items: What was changed, who did a change, when, and why

Possibility to retrieve a specific revision of a configuration item



git



Creating a repository
(not part of the quiz)

```
mkdir my-git-repo
cd my-git-repo
git init --bare
git symbolic-ref HEAD refs/heads/main
```

remote server

Setting up the git environment
(not part of the quiz)

```
git config --global user.name "Your Name"

git config --global user.email “your@email.com"

git config --global pull.rebase false
```

local computer

Cloning a repository

```
git clone <url>
```

local computer

Checking the status of the working copy

```
git status
```

local computer

Pushing changes

```
git add <file>
```
Tells git you want to commit that file

```
git commit –m "commit message”
```
Commits a change on your local repository
(the change is not sent to the remote repository)

```
git push
```
Send the change to the remote repository

local computer

Pulling changes from the remote repository

```
git pull
```

Tells git to retrieve the changes from the remote repositories (e.g., changes pushed by other developers) and merge them in your local working copy

local computer

Undoing changes in the local working copy

```
git checkout .
```

All local changes (not yet "added" with git add) are reverted

```
git reset <file>
```

Undo a previously performed git add <file>

local computer

Retrieving an old version of a file

```
git checkout <commit-id> -- <file>
```

Note: To put it back in the repo, you need the usual add, commit, push.

local computer

Deleting a file from the repository

```
git rm <file>
```

Removes a file both from the local working copy and from the remote repository. It must be followed by git commit and push commands. Use the -r flag for directories.

```
git rm --cached <file>
```

Same as above, but the file is only removed from the remote repository (you still have it on the local working copy)

local computer

Handling conflicts

```
git status
```

To check what is the conflicted <file>

```
Modify the file as needed to solve
the conflict.
```

```
git add <file>
git commit -m "commit message"
git push
```

local computer